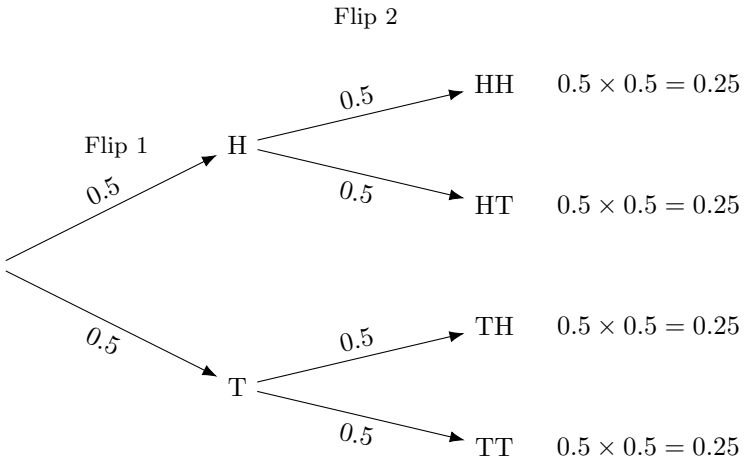
Probability Trees and Relative Risk

A **probability tree** shows every possible outcome of two or more events, one stage at a time.

- Label every branch with its probability. The branches leaving any single point must add up to 1.
- To find the probability of a path (a sequence of outcomes), **multiply** the probabilities along the branches.
- To find the probability of several different paths (e.g. "at least one"), **add** the probabilities of those paths.

Worked Example

A fair coin is flipped twice. Here is the completed probability tree.



Check: the four outcome probabilities add up to $0.25 + 0.25 + 0.25 + 0.25 = 1$. $P(\text{at least one head}) = 1 - P(TT) = 1 - 0.25 = 0.75$.

1 Draw Your Own Probability Trees

For each scenario: draw and label a complete probability tree (including the outcomes and their probabilities at the end of each path), then answer the question(s).

1. A spinner is divided into 3 equal sections: Red, Blue and Green. It is spun twice.

   Draw your tree here.

   Find P(both spins are the same colour). _____

2. A bag contains 4 red marbles and 1 blue marble. A marble is drawn, its colour is noted, and it is **put back** in the bag. A second marble is then drawn.

   Draw your tree here.

   Find P(at least one blue marble is drawn). _____

3. The same bag (4 red, 1 blue) is used again, but this time the first marble is **not** put back before the second marble is drawn.

Draw your tree here.

(a) Find P(both marbles are red). _____

(b) Explain why P(both marbles are blue) must be 0.

_____

4. The probability that it rains on Monday is 0.4. If it rains on Monday, the probability it also rains on Tuesday is 0.7. If it does **not** rain on Monday, the probability it rains on Tuesday is 0.3.

   Draw your tree here.

   (a) Find P(it rains on both days). _____

   (b) Find P(it rains on exactly one of the two days). _____

5. A box contains 10 electronic components, 3 of which are defective. Two components are selected at random, one after the other, **without replacement**, for testing.

   Draw your tree here.

   (a) Find P(both components are defective). _____

   (b) Find P(at least one component is defective). _____

6. In a certain population, 2% of people have a particular disease. A screening test correctly identifies 90% of people who **have** the disease as positive, and correctly identifies 95% of people who do **not** have the disease as negative.

Draw your tree here.

(a) Find P(a randomly chosen person has the disease *and* tests positive). _____

(b) Find P(a randomly chosen person tests positive), using both branches that lead to a positive result.

_____

2 Relative Risk

Relative risk compares how likely an event is in one group compared to another group.

$$\text{Relative risk (RR)} = \frac{\text{risk in Group A}}{\text{risk in Group B}}, \qquad \text{where risk} = \frac{\text{number with the outcome}}{\text{total number in the group}}.$$

If $RR = 1$, the risk is the same in both groups. If $RR > 1$, Group A has a higher risk. If $RR < 1$, Group A has a lower risk.

1. In a drug trial, 400 people took a new medicine and 20 of them experienced side effects. 600 people took a placebo (an inactive dummy pill) and 18 of them experienced side effects.
   (a) Find the risk of side effects for the medicine group. _____
   (b) Find the risk of side effects for the placebo group. _____
   (c) Calculate the relative risk of side effects for the medicine group compared to the placebo group. ____
   (d) Write a sentence interpreting your answer to part (c).

   _____

2. Out of 250 smokers studied, 45 developed a certain lung condition. Out of 750 non-smokers studied, 30 developed the same condition.
   (a) Find the risk of developing the condition for smokers. _____
   (b) Find the risk of developing the condition for non-smokers. _____
   (c) Calculate the relative risk of developing the condition for smokers compared to non-smokers. _____
   (d) Write a sentence interpreting your answer to part (c).

   _____

3. Look back at your probability tree for Question 6 in Section 1 (the disease screening test).
   (a) What is the risk (probability) of testing positive for someone who **has** the disease? _____
   (b) What is the risk (probability) of testing positive for someone who does **not** have the disease? _____
   (c) Calculate the relative risk of testing positive for someone with the disease, compared to someone without the disease.

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